

```
1 import cv2
2 import numpy as np
3
4 def compute_dlt(pts_src, pts_dst):
5     A = []
6     for i in range(4):
7         x, y = pts_src[i][0], pts_src[i][1]
8         u, v = pts_dst[i][0], pts_dst[i][1]
9         A.append([-x, -y, -1, 0, 0, 0, x*u, y*u, u])
10        A.append([0, 0, 0, -x, -y, -1, x*v, y*v, v])
11    A = np.array(A)
12    _, _, V = np.linalg.svd(A)
13    H = V[-1].reshape(3, 3)
14    return H / H[2, 2]
15
16 img = cv2.imread(r"C:\Users\Vishal\Desktop\PMLA HEATWAVES\Picture1.png")
17 (h, w) = img.shape[:2]
18
19 pts_src = np.float32([[0, 0], [w-1, 0], [w-1, h-1], [0, h-1]])
20 pts_dst = np.float32([[0, 0], [w-1, 0], [int(0.9*w), h-1], [int(0.1*w), h-1]])
21
22 H = compute_dlt(pts_src, pts_dst)
23 result = cv2.warpPerspective(img, H, (w, h))
24
25 cv2.imshow("Original", img)
26 cv2.imshow("DLT Transformed", result)
27 cv2.waitKey(0)
28 cv2.destroyAllWindows()
```